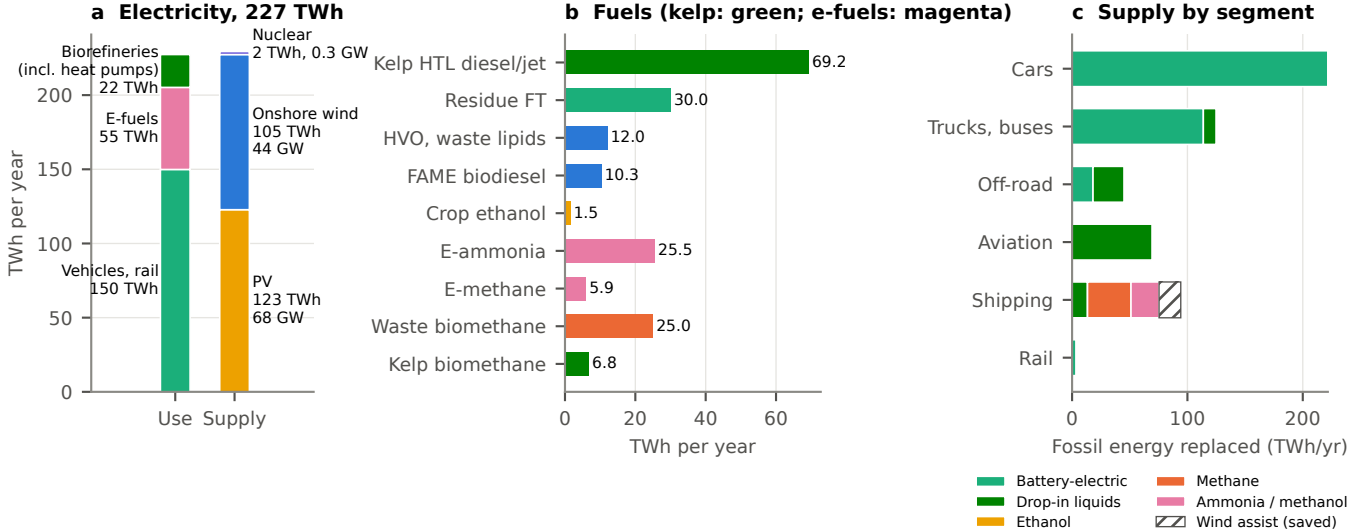
a Electricity, 227 TWh
Biorefineries (incl. heat pumps) 22 TWh
E-fuels 55 TWh
Vehicles, rail 150 TWh
Nuclear 2 TWh, 0.3 GW
Onshore wind 105 TWh 44 GW
PV 123 TWh 68 GW
TWh per year
0
50
100
150
200
Use
Supply
b Fuels (kelp: green; e-fuels: magenta)
Kelp HTL diesel/jet
Residue FT
HVO, waste lipids
FAME biodiesel
Crop ethanol
E-ammonia
E-methane
Waste biomethane
Kelp biomethane
69.2
30.0
12.0
10.3
1.5
25.5
5.9
25.0
6.8
0
20
40
60
TWh per year
c Supply by segment
Cars
Trucks, buses
Off-road
Aviation
Shipping
Rail
0
100
200
Fossil energy replaced (TWh/yr)
Battery-electric
Drop-in liquids
Ethanol
Methane
Ammonia / methanol
Wind assist (saved)